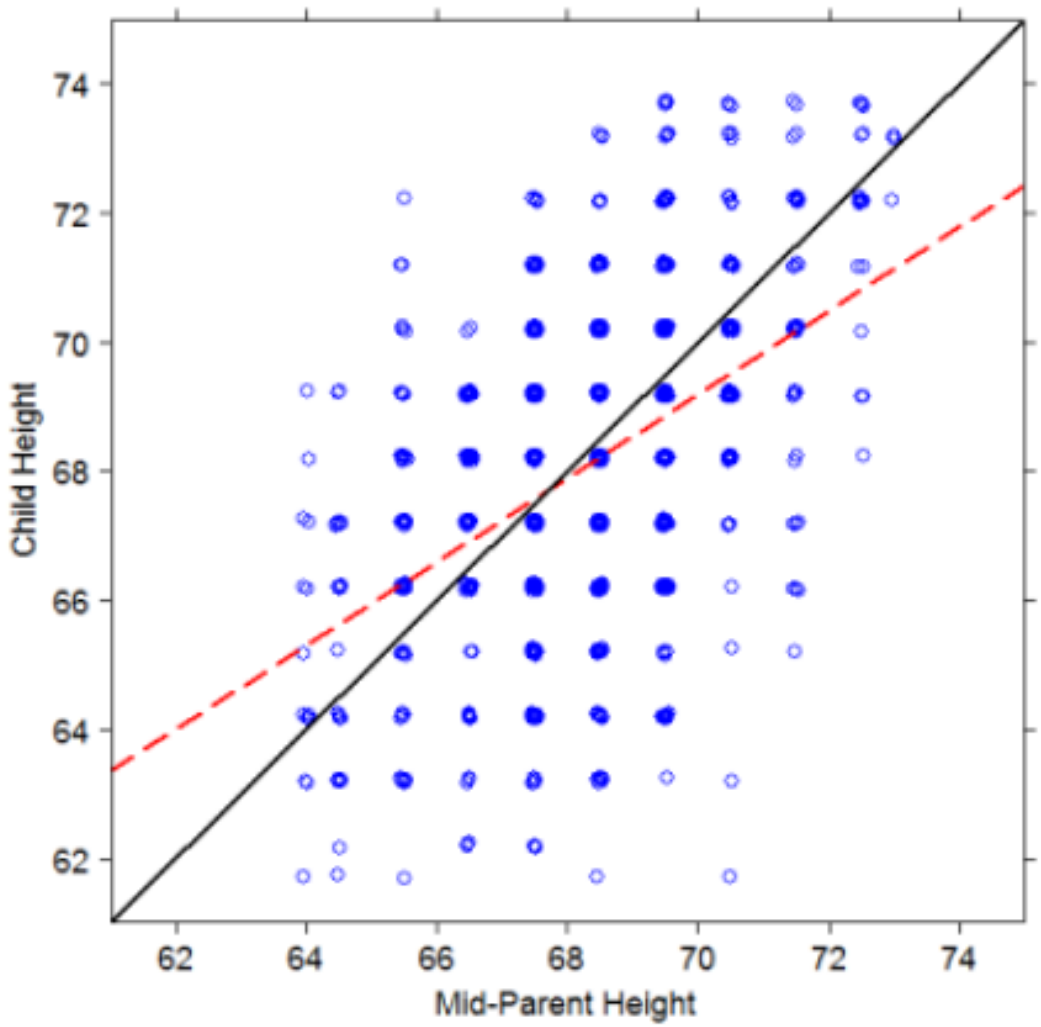
CSE 270 Sports Analytics

Regresion to the mean

Habet Madoyan

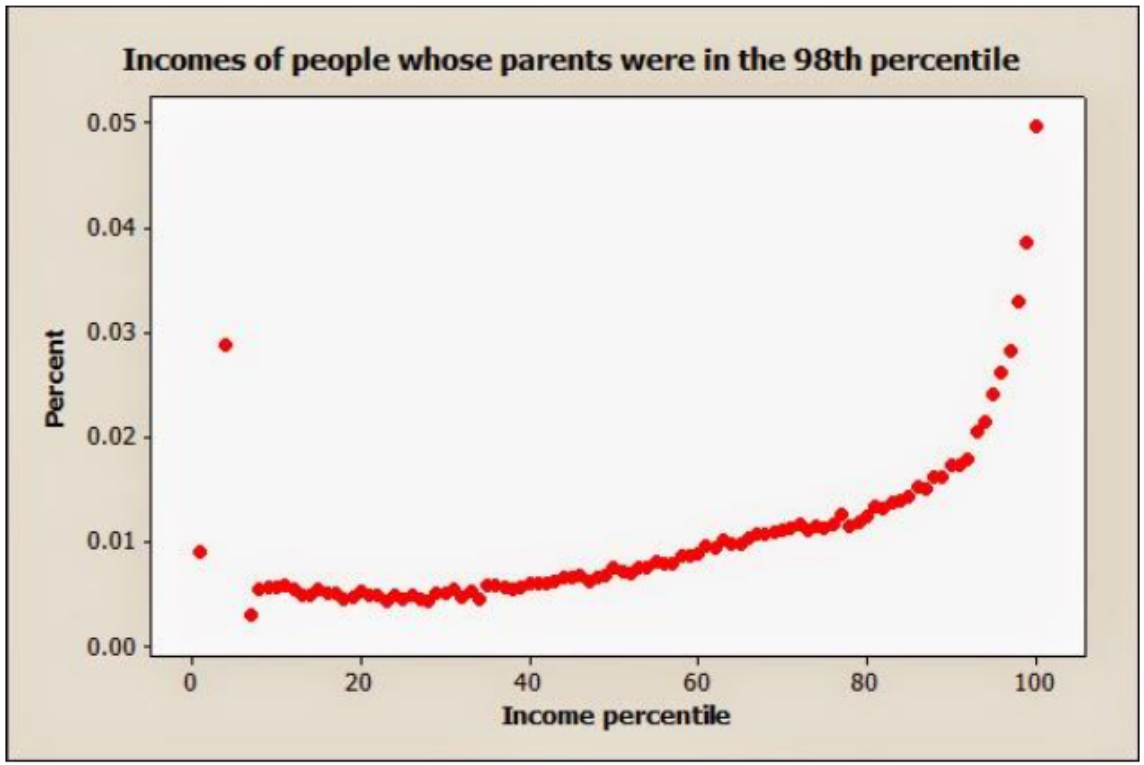
Regression to the mean: Height

Sir Francis Galton proved that the kids of shorter parents have higher height than their parents, and kids of taller parents have shorter height than their parents



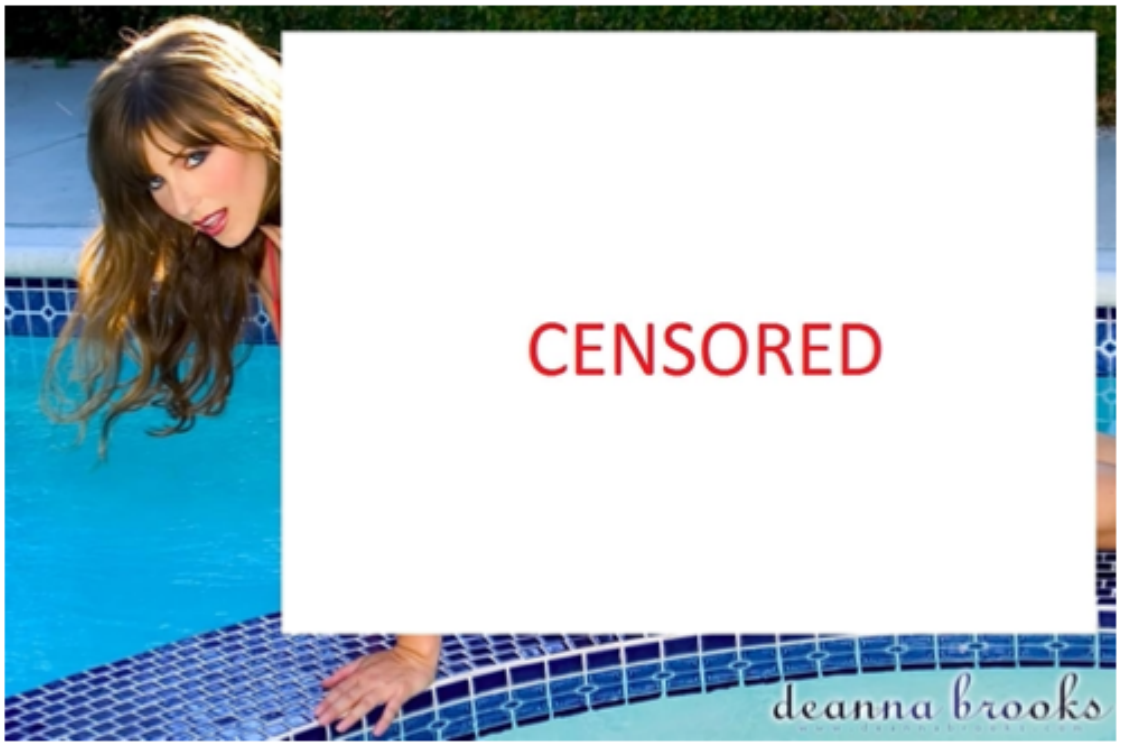
Regression to the mean: Wealth

If your income is at the 98th percentile of the income distribution — that is, you earn more than 98 percent of the population — the best guess is that your children, when they are adults, will be in the 65th percentile.

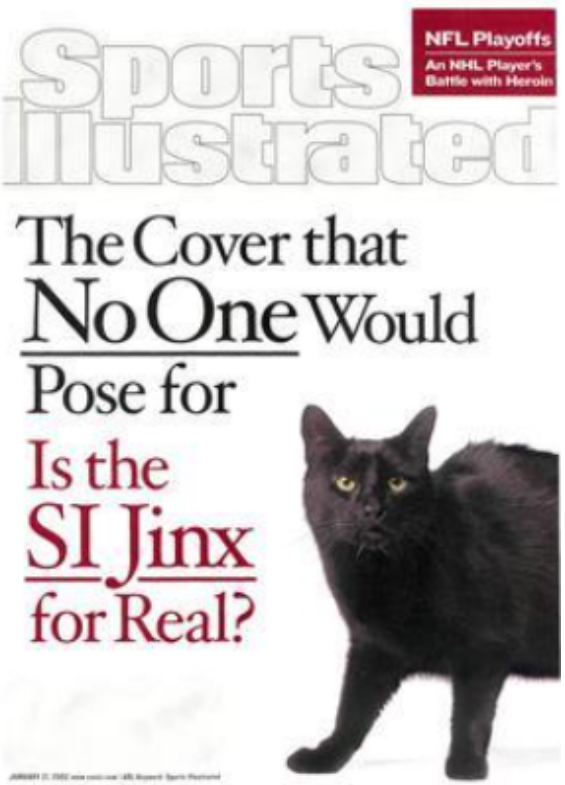


Regression to the mean: Examples

IN 2006, TRADINGMARKETS, a company that helps people trade stocks, asked ten Playboy Playmates to select five stocks each. The idea was to see if they could beat the market. The winner was Deanna Brooks, Playmate of the Month in May 1998. The stocks she picked rose 43.4 percent, trouncing the S& P 500, which gained 13.6 percent, and beating more than 90 percent of the money managers who actively try to outperform a given index. Brooks wasn't the only one who fared well. Four of the other ten Playmates had better returns than the S& P 500 while less than a third of the active money managers did.

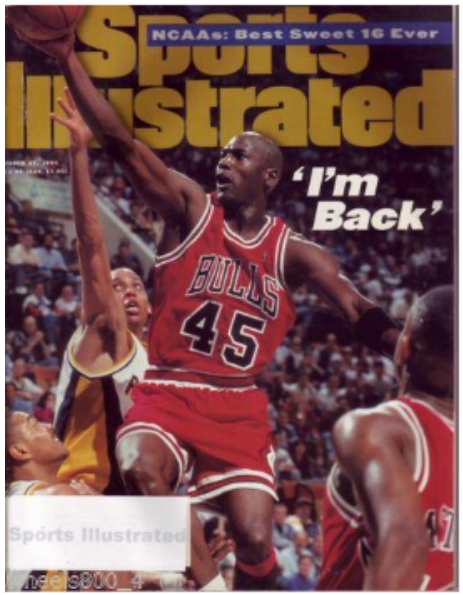


Sports Illustrated cover Jynx



***The Sports Illustrated* cover jinx** is an urban legend that states that individuals or teams who appear on the cover of the Sports Illustrated magazine will subsequently be jinxed (experience bad luck)

Exception



::::

Regression to the mean: Definitions

Regression to the mean is a statistical phenomenon that can make natural variation in repeated data look like real change. It happens when unusually large or small measurements tend to be followed by measurements that are closer to the mean.

Regression to the mean is a technical way of saying that things tend to even out over time.

The term [**Regression to the mean**] is now generally used to label the phenomenon that a variable that is extreme on its first measurement will tend to be closer to the center of the distribution for a later measurement.

True score theory

From the perspective of classical test theory, an examinee's obtained test score (X) is composed of two components, a true score component (T) and an error component (E)

$$X = T + E$$

Skills and Luck: the formula

$$Outcome = Skills + Luck$$

$$Skills = Outcome - Luck$$

How much is the skill and how much is the luck ?

$$Skills = Outcome - Luck$$

$$Var(Skills) = Var(Outcome) - Var(luck)$$

Example: NBA

We will assume that the game outcome has Binomial distribution

$$P(\text{win x games}) = \frac{n!}{x!(n-x)!} p^x * (1-p)^{(n-x)}$$

$$\mu = np$$

$$\sigma^2 = np(1-p)$$

Example: NBA

```
data("nba_misc")
head(nba_misc)
```

```
##   Rk                  Team  Age  W  L PW PL  MOV   SOS  SRS  ORtg  DRtg  Pace
## 1  1        Milwaukee Bucks 26.9 60 22 61 21 8.87 -0.82 8.04 113.8 105.2 103.3
## 2  2  Golden State Warriors 28.4 57 25 56 26 6.46 -0.04 6.42 115.9 109.5 100.9
## 3  3        Toronto Raptors 27.3 58 24 56 26 6.09 -0.60 5.49 113.1 107.1 100.2
## 4  4             Utah Jazz 27.3 50 32 54 28 5.26  0.03 5.28 110.9 105.7 100.3
## 5  5        Houston Rockets 29.2 53 29 53 29 4.77  0.19 4.96 115.5 110.7  97.9
## 6  6 Portland Trail Blazers 26.2 53 29 51 31 4.20  0.24 4.43 114.7 110.5  99.1
##     FTr X3PAr   TSP eFGP_o TOVP_o ORBP_o FT_FGA_o eFGP_d TOVP_d DRBP_d FT_FGA_d
## 1 0.255 0.419 0.583  0.550   12.0   20.8    0.197  0.503   11.5   80.3    0.162
## 2 0.227 0.384 0.596  0.565   12.6   22.5    0.182  0.508   11.7   77.1    0.205
## 3 0.247 0.379 0.579  0.543   12.4   21.9    0.198  0.509   13.1   77.1    0.190
## 4 0.295 0.394 0.572  0.538   13.4   22.9    0.217  0.507   12.4   80.3    0.189
## 5 0.279 0.519 0.581  0.542   12.0   22.8    0.221  0.525   13.4   74.4    0.210
## 6 0.258 0.339 0.568  0.528   12.1   26.6    0.210  0.516   11.0   77.9    0.195
##                     Arena Attendance
## 1            Fiserv Forum     721692
## 2            Oracle Arena     803436
## 3        Scotiabank Arena     812822
## 4 Vivint Smart Home Arena     750546
## 5           Toyota Center     740392
## 6             Moda Center     799345
```

Example: NBA

Every team has played 82 games. Create another variable with win ratio

```
nba_misc$WR<-nba_misc$W/82
```

Lets get the variance of the winning ratio- This is the variance for outcome

```
var(nba_misc$WR)
```

```
## [1] 0.02151839
```

Let us look at the variance for the luck. Here we assume that the chance of winning for each team is the same 0.5.

```
0.5*0.5/82
```

```
## [1] 0.00304878
```

For variance of the luck (proportion p), we use the following formula

$$\frac{p(1-p)}{n}$$

Example: NBA

Var(Skills) = Var(Outcome) - Var(Luck)

Var(Skills) = 0.02151839 - 0.00304878

Var(Skills) = 0.01846961

The paradox fo Skill:
More skills means luck is more important

Assuming that the variance in luck stays constant

Example: NBA

Var(Skills) = Var(Outcome) - Var(Luck)

Var(Skills) = 0.01861577 - 0.00304878

Var(Skills) = 0.01556699

This analysis allows us to look at the ratio of variance (luck) to variance (observed) in order to determine the contribution of luck.

$$\frac{0.00304878}{0.01556699} = 19\%$$

Paradox of skill

The less variation in skill there is, the more important luck is.

Skill or luck ?

Manchester United vs Bayern Munich UCL Final

1999

https://youtu.be/YnecJa5satA?t=112

Manchester United vs Bayern Munich, UCL final 1999

Overall
Box-Score Statistics

Statistic	MU	Bayern
Goals Scored	2	1
Total Shots	15	15
Shots on target	9	7
Ball possession	53%	47%
Corner kicks	12	7
Fouls committed	11	10
Offsides	6	8
Yellow cards	0	1
Red cards	0	0

So what ?

The contribution of luck in some professional sports leagues

League	Contribution of luck	Five seasons ended
National Basketball Association	12%	2011
Premier League	31%	2011
Major League Baseball	34%	2011
National Football League	38%	2011
National Hockey League	53%	2012

Source: Analysis by author.

Mauboussin, Michael J.. The Success Equation: Untangling Skill and Luck in Business, Sports, and Investing. Harvard Business Review Press.

Curse of the Bambino



- In 1919 Red Sox sold their star player, Babe Ruth to New York Yankees
- Red Sox was not able to win World Series for the next 86 years, despite of being a favorite in many final games

Regression to the mean

Construct the variable DF of difference

```
data(nba_east)
data(nba_west)
nba_st <- rbind(nba_east,nba_west)


nba_st$PW <- nba_st$PS^14.105/(nba_st$PS^14.105+nba_st$PA^14.105)
nba_st$G <- nba_st$W+nba_st$L
nba_st$PW_G <- round(nba_st$PW*nba_st$G)

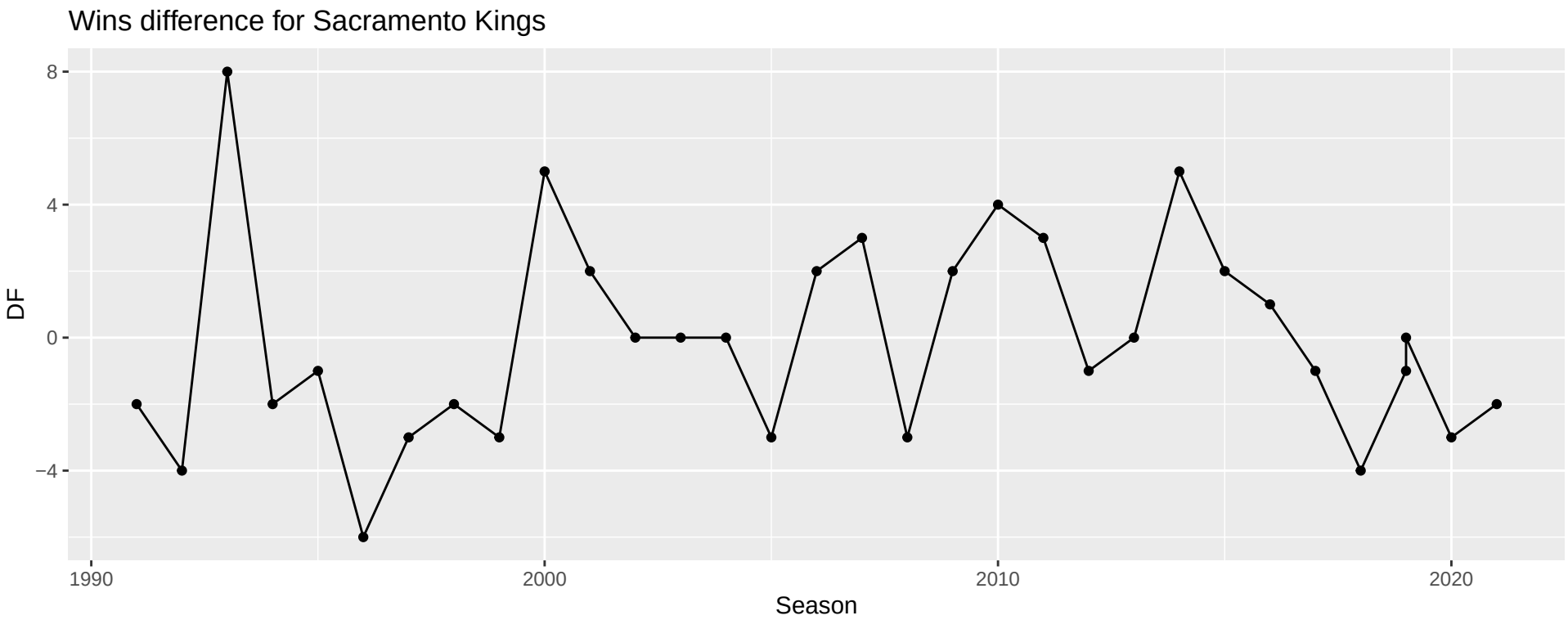
nba_st$DF <- nba_st$PW_G-nba_st$W
```

Regression to the mean

Let's plot for one team

```
nba_st$DF <- nba_st$PW_G-nba_st$W

nba_st %>% filter(Team=="Sacramento Kings") %>%
ggplot(aes(x=Season, y=DF))+geom_point()+geom_line()+
  ggtitle('Wins difference for Sacramento Kings')
```



Regression to the mean

```
means <- nba_st %>% group_by(Team) %>% summarise(mean=mean(DF))
head(means)
```

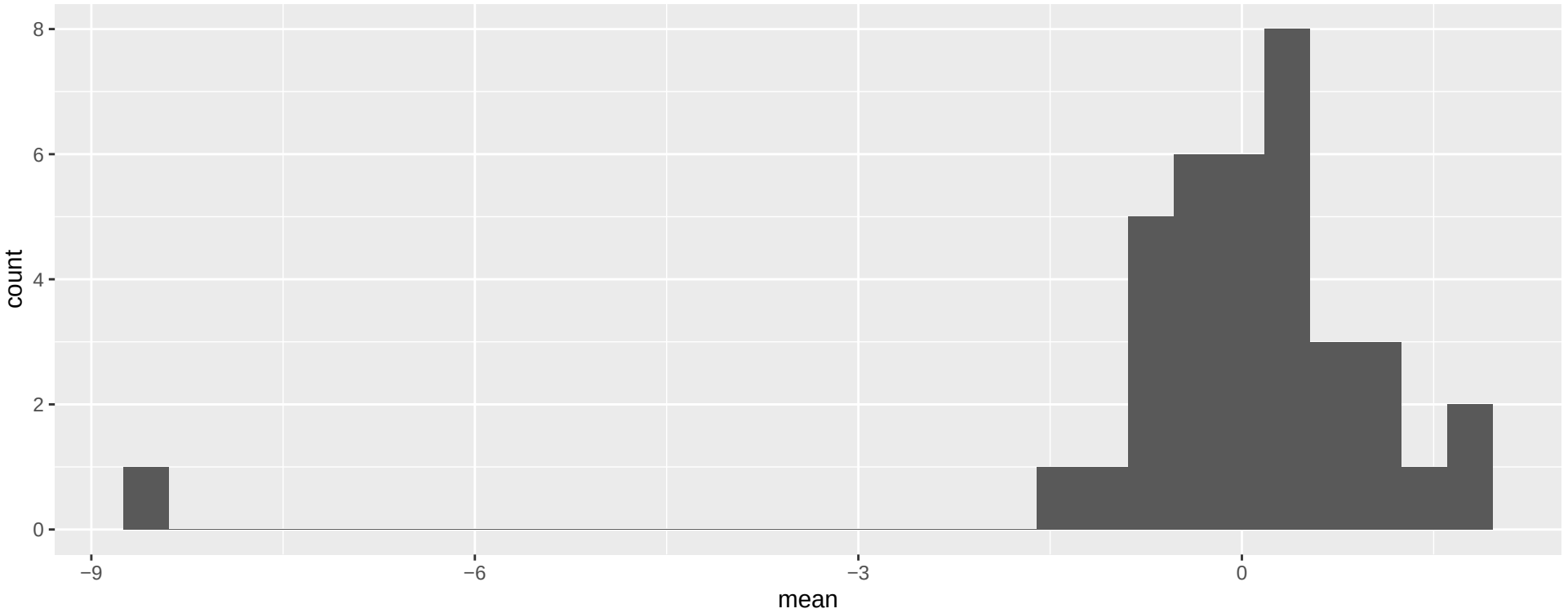
```
## # A tibble: 6 x 2
##   Team                 mean
##   <chr>               <dbl>
## 1 Atlanta Hawks      -0.594
## 2 Boston Celtics      0.688
## 3 Brooklyn Nets      -1
## 4 Charlotte Bobcats  -0.5
## 5 Charlotte Hornets  -0.6
## 6 Chicago Bulls      -0.781
```

Average difference by teams (You need dplyr here)

Regression to the mean

The histogram of the averages

```
ggplot(means, aes(x=mean))+geom_histogram()
```



Competitive Balance in Sports

Competitive Balance in Sports

Competitive Balance

Definition from economics:

A market situation where no business is too big or has an unfair advantage. Source: http://lexicon.ft.com/Term?term=competitive-balance

Definition in Sports:

The degree of uncertainty in the match outcome or the final league standings.

Higher is the Competitive Balance, higher is the uncertainty

Competitive Balance: Dimensions

Long-run

Distribution of championships over time

Seasonal

Within season

Short-run

Match uncertainty Outcome of contest

Measuring Seasonal Dimensions

- Statistical measures
  - Range, variance or deviation
- Standard deviation (SD)
  - Dispersion from mean
  - If SD is low, then CB is high
- CB is often measured by
  - SD of actual wins $\div$ idealized SD
  - As measure increases, competitive balance worsens

Seriea: Measuring balance with standard deviation

```
data("seriea_st")
colnames(seriea_st)
```

```
##  [1] "Rank"   "Team"   "M"      "W"      "D"      "L"      "Goals"  "Diff"
##  [9] "Pt"     "Season"
```

The dataset

```
head(seriea_st)
```

```
##   Rank           Team  M  W  D L Goals Diff Pt Season
## 1    1       AC Milan 34 21 10 3 60:24   36 73   1996
## 2    2       Juventus 34 19  8 7 58:35   23 65   1996
## 3    3     Lazio Roma 34 17  8 9 66:38   28 59   1996
## 4    4 ACF Fiorentina 34 17  8 9 53:41   12 59   1996
## 5    5        AS Roma 34 16 10 8 51:34   17 58   1996
## 6    6       Parma AC 34 16 10 8 44:31   13 58   1996
```

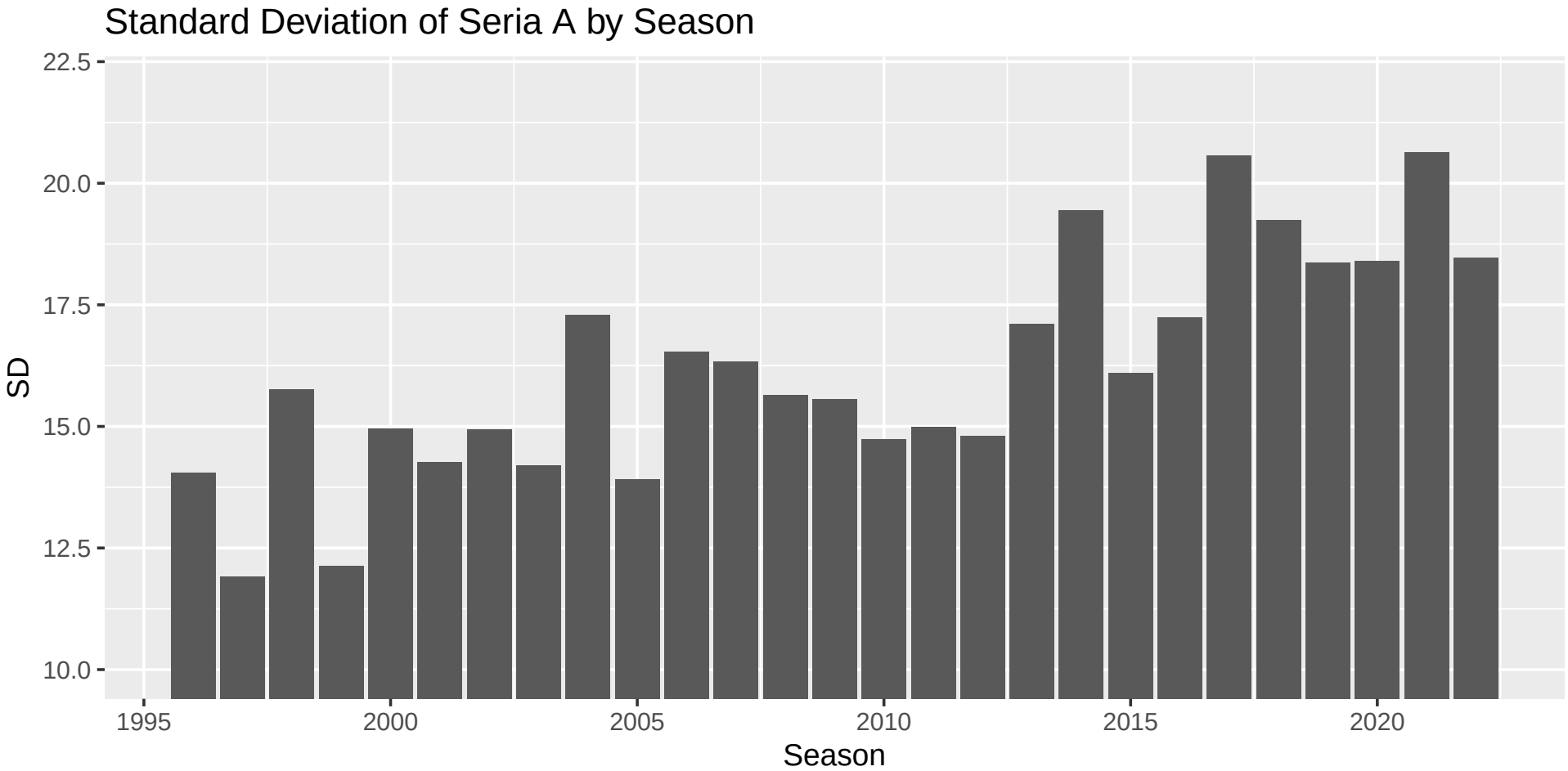
Standard Deviations: Serie A

```
seriea_s<-seriea_st %>%
  group_by(Season) %>%
  summarise(SD=sd(Pt))
seriea_s
```

```
## # A tibble: 27 x 2
##    Season    SD
##     <dbl> <dbl>
##  1   1996  14.0
##  2   1997  11.9
##  3   1998  15.8
##  4   1999  12.1
##  5   2000  15.0
##  6   2001  14.3
##  7   2002  14.9
##  8   2003  14.2
##  9   2004  17.3
## 10   2005  13.9
## # ... with 17 more rows
## # i Use `print(n = ...)` to see more rows
```

Standard Deviations: Serie A

```
ggplot(seriea_s, aes(Season, SD))+geom_bar(stat="identity")+
  ggtitle("Standard Deviation of Seria A by Season")+
  coord_cartesian(ylim=c(10,22))
```



NFL

NFL data

```
data(nfl_st)
summary(nfl_st)
```

```
##       Tm                  W                L                T
##  Length:763         Min.   : 0.000   Min.   : 0.000   Min.   :-1.00000
##  Class :character   1st Qu.: 6.000   1st Qu.: 6.000   1st Qu.: 0.00000
##  Mode  :character   Median : 8.000   Median : 8.000   Median : 0.00000
##                     Mean   : 8.005   Mean   : 8.005   Mean   :-0.01048
##                     3rd Qu.:10.000   3rd Qu.:10.000   3rd Qu.: 0.00000
##                     Max.   :16.000   Max.   :16.000   Max.   : 1.00000
##       Wpct              PF              PA              PD
##  Min.   :0.0000   Min.   :161.0   Min.   :165.0   Min.   :-261.0
##  1st Qu.:0.3750   1st Qu.:300.5   1st Qu.:312.5   1st Qu.: -75.0
##  Median :0.5000   Median :349.0   Median :350.0   Median :   1.0
##  Mean   :0.5002   Mean   :352.8   Mean   :352.8   Mean   :   0.0
##  3rd Qu.:0.6250   3rd Qu.:399.5   3rd Qu.:393.0   3rd Qu.:  73.5
##  Max.   :1.0000   Max.   :606.0   Max.   :519.0   Max.   : 315.0
##       MoV                  SoS                SRS
##  Min.   :-16.300000   Min.   :-5.900000   Min.   :-1.74e+01
##  1st Qu.: -4.700000   1st Qu.:-1.100000   1st Qu.:-4.40e+00
##  Median :  0.100000   Median : 0.000000   Median : 1.00e-01
##  Mean   : -0.001704   Mean   : 0.001311   Mean   : 5.24e-04
##  3rd Qu.:  4.600000   3rd Qu.: 1.200000   3rd Qu.: 4.50e+00
##  Max.   : 19.700000   Max.   : 4.300000   Max.   : 2.01e+01
##       OSRS               DSRS              Season         Conf
##  Min.   :-1.23e+01   Min.   :-9.800000   Min.   :1998   Length:763
##  1st Qu.:-3.20e+00   1st Qu.:-2.350000   1st Qu.:2004   Class :character
##  Median :-1.00e-01   Median : 0.100000   Median :2010   Mode  :character
##  Mean   :-1.31e-04   Mean   :-0.001573   Mean   :2010
##  3rd Qu.: 2.80e+00   3rd Qu.: 2.500000   3rd Qu.:2016
##  Max.   : 1.59e+01   Max.   : 9.800000   Max.   :2021
##      Div
##  Length:763
##  Class :character
##  Mode  :character
```

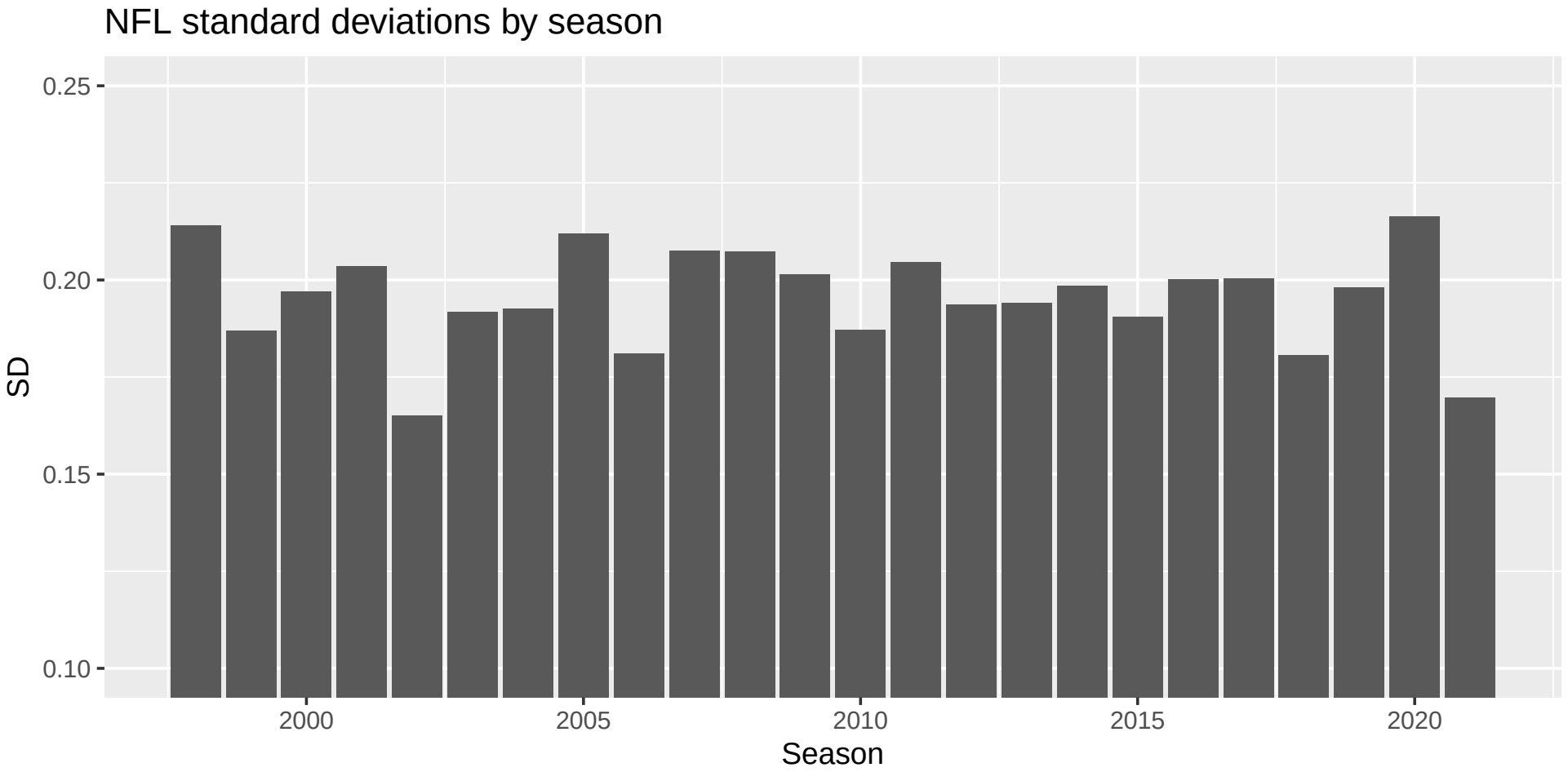
NFL data: SD

```
nfl_s<-nfl_st %>%
  group_by(Season) %>%
  summarise(SD=sd(Wpct))
nfl_s
```

```
## # A tibble: 24 x 2
##    Season    SD
##     <int> <dbl>
##  1   1998 0.214
##  2   1999 0.187
##  3   2000 0.197
##  4   2001 0.203
##  5   2002 0.165
##  6   2003 0.192
##  7   2004 0.192
##  8   2005 0.212
##  9   2006 0.181
## 10   2007 0.208
## # ... with 14 more rows
## # i Use `print(n = ...)` to see more rows
```

NFL data: SD

```
ggplot(nfl_s, aes(Season, SD))+geom_bar(stat="identity")+
  ggtitle("NFL standard deviations by season")+
  coord_cartesian(ylim=c(0.1,0.25))
```



StDev approach: the drawbacks

- Depends on how the game is played and how the points are distributed
- Cannot be used to compare different sports

Measuring Seasonal Dimensions

There are several drawbacks of using standard deviation as a measure of competitive balance.

n and the number of the games played are not always constant over time or same for different sports and leagues.

To deal with this problem, σ is benchmarked against standard deviation that describes dispersion for the league with ideal competitive balance.

Measuring Seasonal Dimensions

Lets assume we are dealing with a game, where the match outcome is either win or loose.

As previously, we will define w (wins ratio) as a number of games a team has won, divided by the number of games it has played.

The standard deviation of the wins ratio will be defined as

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (w_i - 0.5)^2}{n}}$$

n - number of teams

w_i - winning ration of team i

Measuring Seasonal Dimensions

Ideally, each team will have 0.5 probability of winning any game (ignoring home-field advantage). Then the ideal standard deviation will be calculated with the following formula

$$\sqrt{\frac{0.5*(1-0.5)}{m}} = \frac{0.5}{\sqrt{m}}$$

Where m is the number of games played by each team.

This is also called Noll-Scully number

Measuring Seasonal Dimensions

Now we can define relative measure of competitive balance

$$\sigma_{\text{relative}} = \frac{\sigma}{0.5/\sqrt{m}}$$

This is also called Noll-Scully number

Measuring Seasonal Dimensions: Noll-Scully number

- Ideally, Noll-Scully number will be equal to 1
- Higher is the number, lower is Competitive Balance
- Higher is the number, lower is Competitive Balance
- Closer is the number to 1, higher is Competitive Balance

Measuring Seasonal Dimensions: Noll-Scully number

The Noll-Scully compares what a league actually looks like to what it would ideally look like. In a perfect world — i.e. one where actual balance matches the ideal — the metric would return a 1.0, meaning the league is balanced.

If the number is above 1.0 it means the teams are further away in wins than we would expect given the ideal.

If the number is below 1.0 — which is very rare case— it means the teams are closer in wins than we would expect.

Noll-Scully number: Calculations

Noll-Scully number

```
#Ideal StDev
id_s <- 0.5/sqrt(16)
# Win percentage StDev
sd_wpct <- sd(nfl_st$Wpct)
# Noll-Scully number
sd_wpct/id_s
```

```
## [1] 1.543418
```

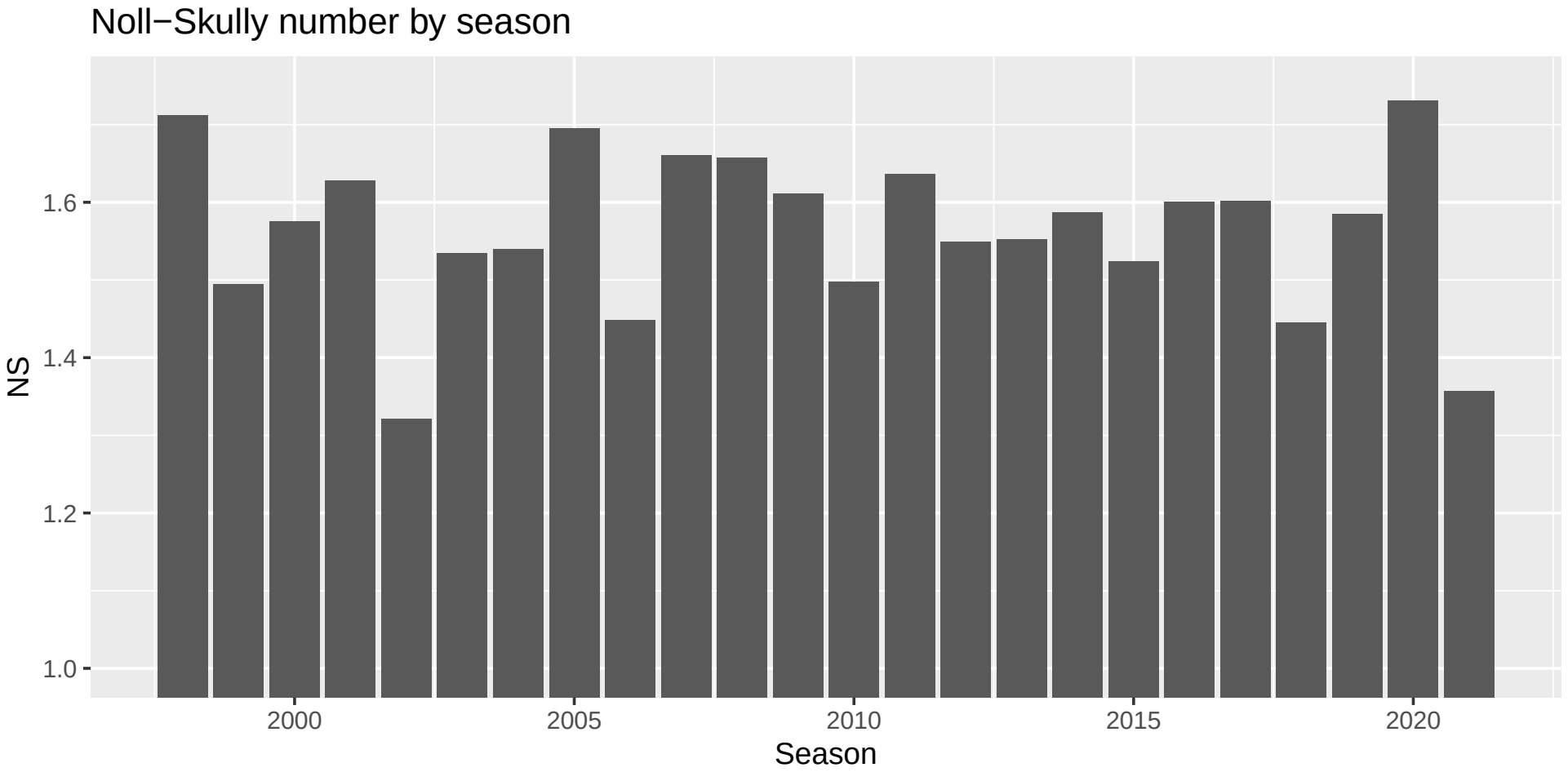
Calculating season-by season

```
nfl_s<-nfl_st %>%
  group_by(Season) %>%
  summarise(NS=sd(Wpct)/id_s)
nfl_s
```

```
## # A tibble: 24 x 2
##    Season    NS
##     <int> <dbl>
##  1   1998  1.71
##  2   1999  1.49
##  3   2000  1.58
##  4   2001  1.63
##  5   2002  1.32
##  6   2003  1.53
##  7   2004  1.54
##  8   2005  1.70
##  9   2006  1.45
## 10   2007  1.66
## # ... with 14 more rows
## # i Use `print(n = ...)` to see more rows
```

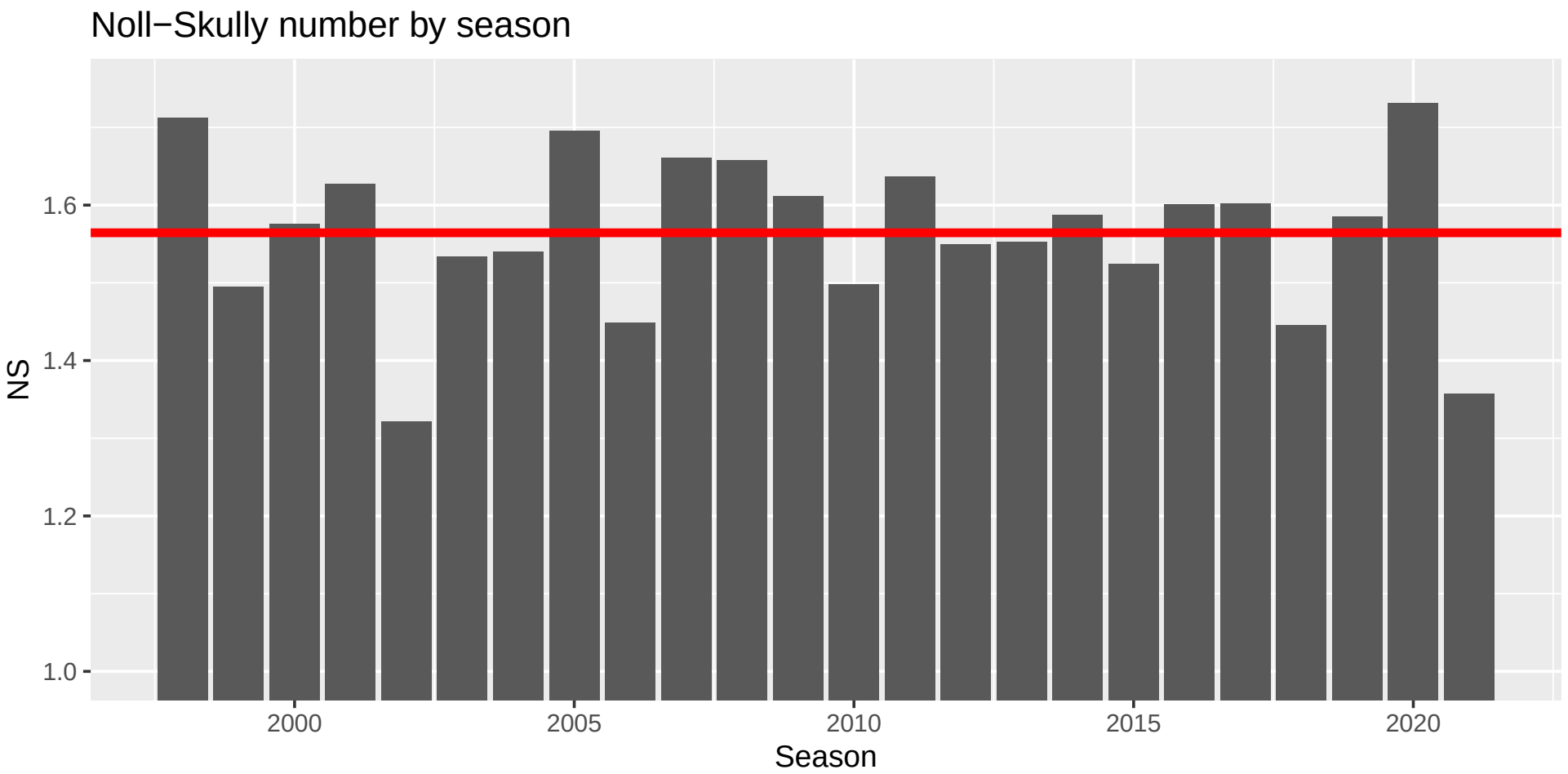
Plotting results

```
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  coord_cartesian(ylim=c(1,1.75))+
  ggtitle("Noll-Skully number by season")
```



Measuring Seasonal Dimensions: Noll-Scully number

```
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  ggtitle("Noll-Skully number by season")+
  coord_cartesian(ylim=c(1,1.75))+
  geom_hline(yintercept=mean(nfl_s$NS), col="red", size=1.5)
```



Previous studies

Twenty Years of Competitive Balance in North America

North American Sports League	Years	Average Noll-Scully Measure of Competitive Balance
National Basketball Association	1985–86 to 2004–05	2.86
American League	1986–2005	1.78
National League	1986–2005	1.76
National Hockey League	1984–85 to 2003–04	1.70
National Football League	1986–2005	1.49

Source: **Berri, D. J., Schmidt, M. B., & Brook, S. L. (2008). *The wages of wins: taking measure of the many myths in modern sport*. Stanford, CA: Stanford University Press.**

Concentration indexes

Concentration indexes

Measuring Seasonal dimensions: Concentration of points

Concentration Index for points

$$C5 = \frac{\text{Number of points earned by top N clubs}}{\text{Total number of points earned by all clubs}}$$

Measuring Seasonal dimensions: Concentration of points

Concentration index for points

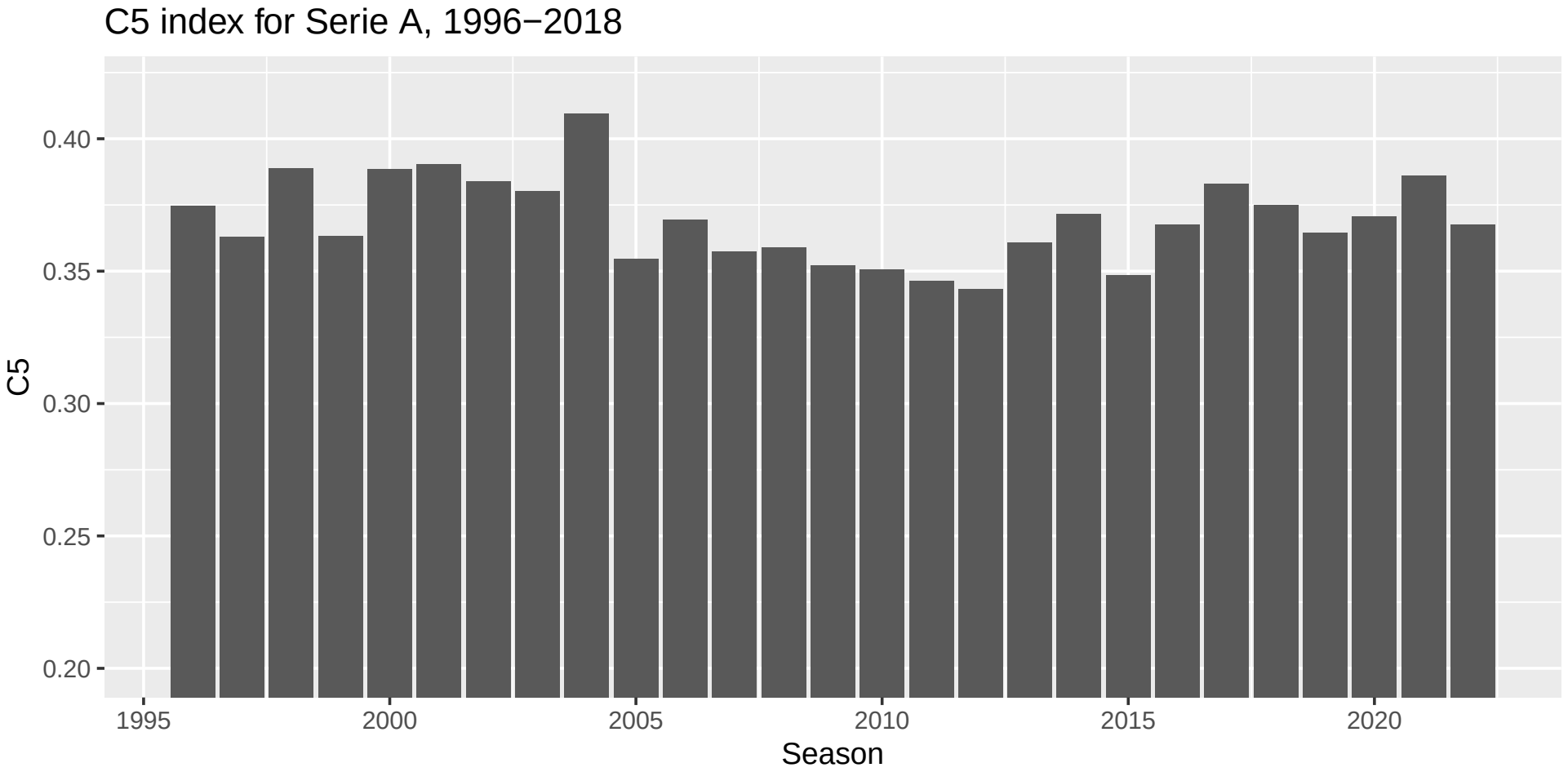
```
top5<- seriea_st %>%
    group_by(Season) %>%
    filter(Rank<6) %>%
    summarize(TopP=sum(Pt))

all<-seriea_st %>%
    group_by(Season) %>%
    summarize(Pt=sum(Pt))

C5<-data.frame(Season=top5$Season,C5=top5$TopP/all$Pt)
```

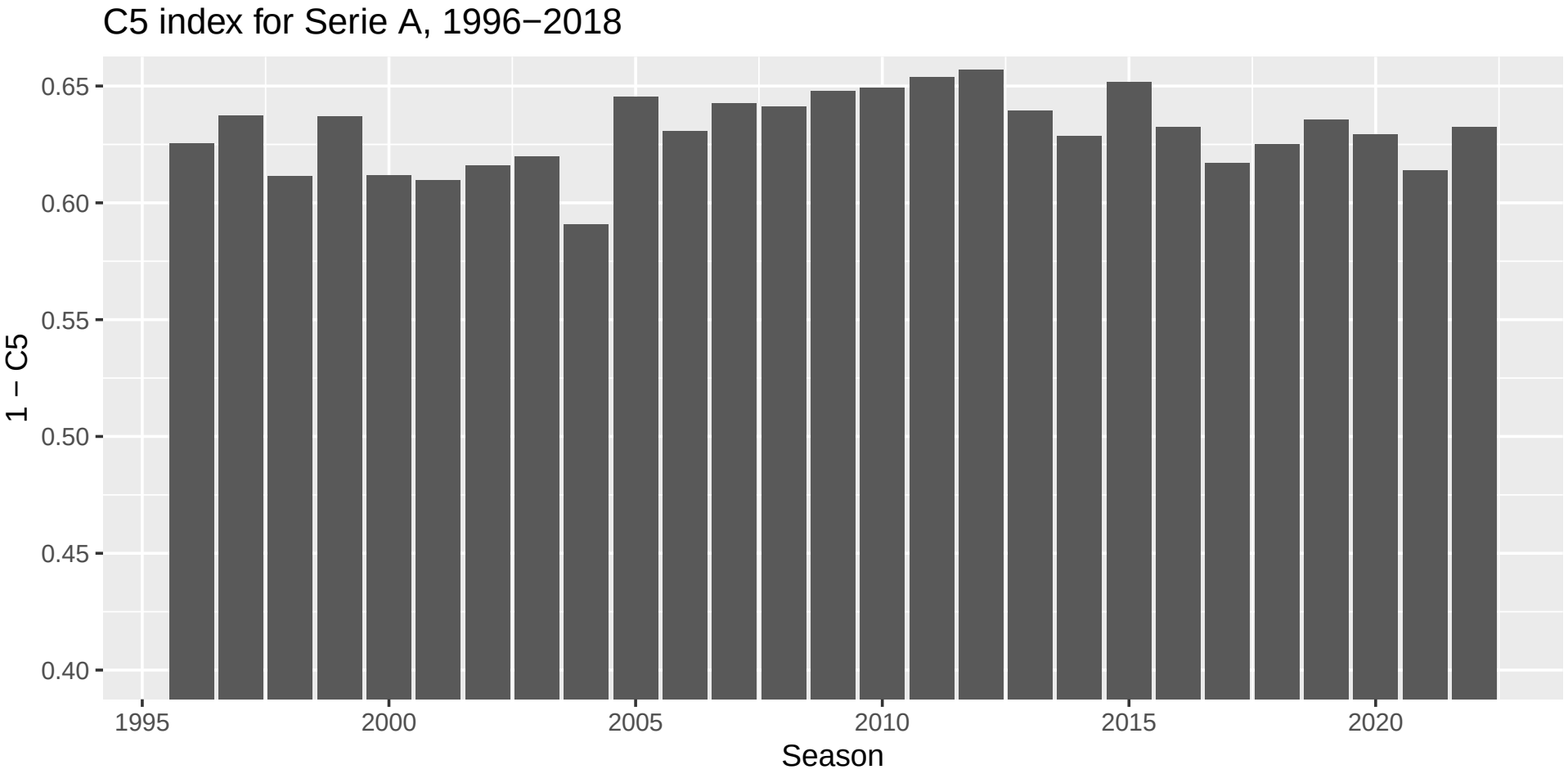
Measuring Seasonal dimensions: Competitive imbalance

```
ggplot(C5, aes(Season, C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2018")+
  coord_cartesian(ylim=c(0.2,0.42))
```



Measuring Seasonal dimensions: Competitive Balance

```
ggplot(C5, aes(Season, 1-C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2018")+
  coord_cartesian(ylim=c(0.4,0.65))
```



Measuring Long Run dimensions

Concentration Index

Proportion of championships (or points) by top n teams

e.g. CI_5 : dominance of top 5 teams

Measuring Long Run dimensions

How many times did every team become a champion?

```
champ_c = seriea_st %>%
  filter(Rank ==1) %>% group_by(Team) %>%
  summarize(champ=n()) %>% arrange(desc(champ))

champ_c
```

```
## # A tibble: 5 x 2
##   Team        champ
##   <chr>       <int>
## 1 Juventus       16
## 2 AC Milan        5
## 3 Inter           5
## 4 AS Roma         1
## 5 Lazio Roma      1
```

CI4 for championship

```
100*sum(champ_c[1:4,2])/sum(champ_c$champ)
```

```
## [1] 96.42857
```

If you take CI5 the concentration index is going to be 100%

Herfindahl-Hirschman Index - HHI

Herfindahl-Hirschman Index - HHI

Herfindahl-Hirschman Index - HHI

Definition comes from economics

The Herfindahl-Hirschman index (HHI) is a commonly accepted measure of market concentration. It is calculated by squaring the market share of each firm competing in a market, and then summing the resulting numbers

Read more: http://www.investopedia.com/terms/h/hhi.asp#ixzz4t8sTjDns

Herfindahl-Hirschman Index - HHI

$$HHI = \sum_{i=1}^{n} w_i^2$$

Simple case: 2 firms in the market

- Highly competitive industry, both firms have the same market share

$HHI = \sum_{i=1}^{2} w_i^2 = 0.5^2 + 0.5^2$

- Monopoly, 1 firm has 100%

$HHI = \sum_{i=1}^{2} w_i^2 = 1^2 + 0^2$

Herfindahl-Hirschman Index - HHI

$$HHI = \sum_{i=1}^{n} w_i^2$$

lower and upper bounds

$$1/n \leq HHI \leq 1$$

Herfindahl-Hirschman Index – in sports

Several variations of the index exists.

$$HHI = \sum_{i=1}^{n} \left(\frac{p_i}{\sum_{i=1}^{n} p_i} \right)^2$$

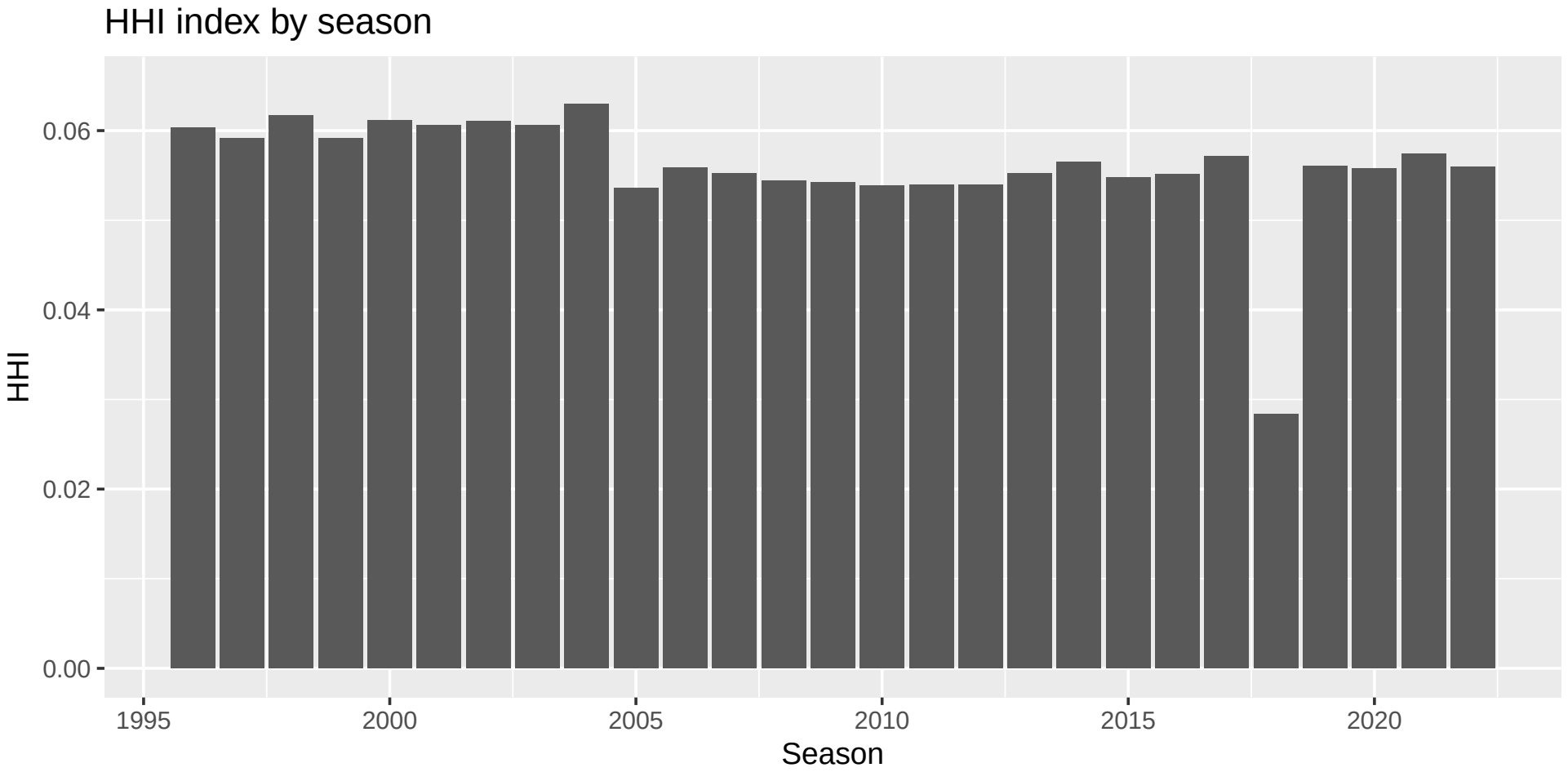
p_i is the number of the points the team has earned.

Doing it in R: Serie A

```
HHI = seriea_st %>%
        group_by(Season) %>%
        mutate(Perc=Pt/sum(Pt))%>%
        summarise(HHI = sum(Perc^2))
```

Doing it in R: Serie A

```
ggplot(HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  ggtitle("HHI index by season")+
  coord_cartesian(ylim=c(0,0.065))
```



Doing it in R: Serie A

Number of the teams per season

```
seriea_st %>%
      group_by(Season) %>%
      summarise(count=n())
```

```
## # A tibble: 27 x 2
##    Season count
##     <dbl> <int>
##  1   1996    18
##  2   1997    18
##  3   1998    18
##  4   1999    18
##  5   2000    18
##  6   2001    18
##  7   2002    18
##  8   2003    18
##  9   2004    18
## 10   2005    20
## # ... with 17 more rows
## # i Use `print(n = ...)` to see more rows
```

Adjusted HHI

Increasing number of the teams affects lower bound (in some cases upper bound as well)

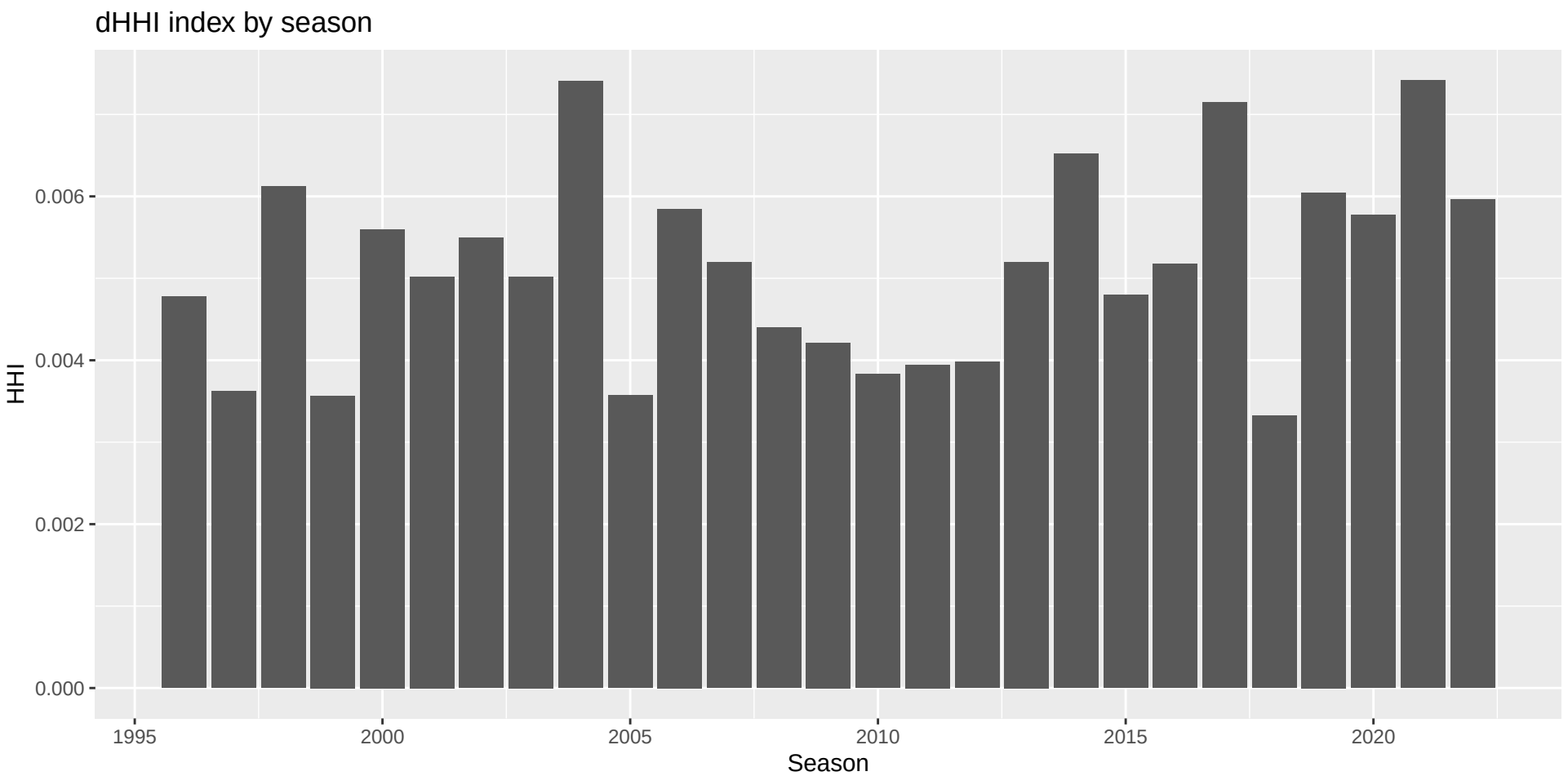
$$dHHI = \sum_{i=1}^{n} w_i^2 - 1/n$$

Adjusted HHI

```
dHHI<-seriea_st %>%
      group_by(Season) %>%
      mutate(Perc=Pt/sum(Pt)) %>%
      summarise(HHI=(sum(Perc^2)-1/n()))
```

dHHI

```
ggplot(dHHI, aes(Season, HHI))+geom_bar(stat="identity")+
  ggtitle("dHHI index by season")
```



HHI defined by Depken

$$HHI = \sum_{i=1}^{n} \left(\frac{2 * wins_i}{N * G_i}\right)^2$$

$wins_i$- *number of wins by each team*

N - *Number of teams*

G_i - *number of games played*

Assumptions

- No ties
- Number of games played by each is an even number

Source: Free-Agency and the Competitiveness of Major League Baseball, CRAIG A. DEPKEN, II, Review of Industrial Organization, Vol. 14, No. 3 (May 1999), pp. 205-217

HHI defined by Depken

$$HHI = \frac{4}{N^2 G^2} \sum_{i=1}^{n} (wins_i)^2$$

Source: Free-Agency and the Competitiveness of Major League Baseball, CRAIG A. DEPKEN, II, Review of Industrial Organization, Vol. 14, No. 3 (May 1999), pp. 205-217

Major League of Baseball: dataset

```
str(mlb_standings)
```

```
## 'data.frame':    690 obs. of  10 variables:
##  $ Season: int  1999 1999 1999 1999 1999 1999 1999 1999 1999 1999 ...
##  $ Lig   : chr  "AL" "NL" "NL" "AL" ...
##  $ Div   : chr  "W" "W" "E" "E" ...
##  $ Team  : chr  "Anaheim Angels" "Arizona Diamondbacks" "Atlanta Braves" "Baltimo
##  $ W     : int  70 100 103 78 94 75 67 96 97 72 ...
##  $ L     : int  92 62 59 84 68 86 95 67 65 90 ...
##  $ G     : int  162 162 162 162 162 162 162 163 162 162 ...
##  $ R     : int  711 908 840 851 836 777 747 865 1009 906 ...
##  $ RA    : int  826 676 661 815 718 870 920 711 860 1028 ...
##  $ Rank  : int  4 1 1 4 2 2 6 2 1 5 ...
```

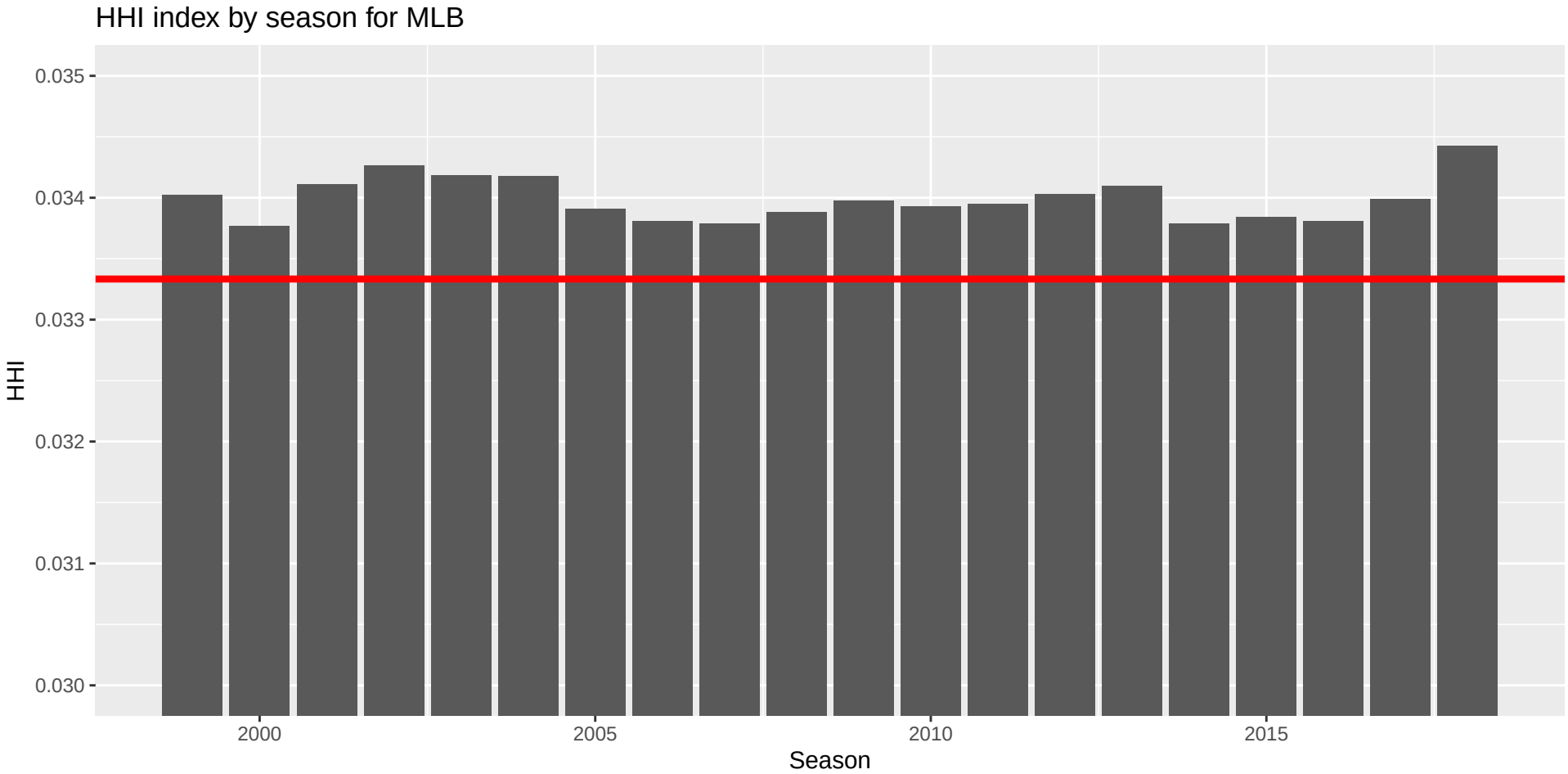
MLB

- Each team plays 162 games per regular season
  - small violations occur, 160,161,163
  - We will assume the number of games to be 162
- 30 teams since 1999

```
mlb_HHI <- mlb_standings %>%
      filter(Season < "2019") %>%
      group_by(Season) %>%
      summarise(HHI=sum(W^2)*(4/(30^2*162^2)))
```

HHI in MLB

```
ggplot(mlb_HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  geom_hline(yintercept = 1/30, col="red", size=1.5)+
  coord_cartesian(ylim=c(0.03,0.035))+
  ggtitle("HHI index by season for MLB")
```



Correlating imbalance measures with other variables

- Leading indicators: input indicators

  Why is there a imbalance ?

  Salary gap ? Macro, socio-economic variables?

- Lagging indicators: output indicators
  - What is affected by the change in imbalance ?
  - Attendance?
  - Teams revenue ?

MLB: Salaries

Leading indicators: SalarY

```
data(mlb_salary)
str(mlb_salary)
```

```
## 'data.frame':    638 obs. of  6 variables:
##  $ Team       : chr  "New York Yankees" "Texas Rangers" "Atlanta Braves" "Clevela
##  $ Opening.day: num  88130709 81301598 75065000 73857962 71720000 ...
##  $ Current    : num  88130709 81301598 75065000 73857962 71720000 ...
##  $ Avg        : num  3147525 2803503 2345781 2051610 2173333 ...
##  $ Median     : num  2008333 2225000 975000 670000 1500000 ...
##  $ Season     : num  1999 1999 1999 1999 1999 ...
```

MLB: Salaries

Aggregate data by year

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
      summarise(Tot.salary=sum(Current))
```

```
cor(mlb_s$Tot.salary, mlb_HHI$HHI)
```

```
## [1] -0.009126879
```

MLB: Salaries

What about the standard deviation of the salary ?

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
      summarise(Tot.salary=sum(Current), SD=sd(Current))
```

```
cor(mlb_s$SD, mlb_HHI$HHI)
```

```
## [1] -0.1180002
```

MLB: Salaries

HHI for salary

```
mlb_HHI_s<-mlb_salary %>%
        group_by(Season) %>%
        mutate(Perc=Current/sum(Current)) %>%
        summarise(HHI_S=sum(Perc^2))
```

Correlations for HHI_S and HHI

```
cor(mlb_HHI_s$HHI_S, mlb_HHI$HHI)
```

```
## [1] 0.06302969
```

Short run competitiveness

Short run competitiveness

Short run competitiveness

- We are looking at how predictable are individual games.
- We can do it by using some machine learning or simpler algorithms (like Elo) to predict the probability of game outcome then calculate the prediction error – High error will indicate higher uncertainty, thus higher competitiveness.
- However, here we will look at the bookmaker's odds as a predicted probabilities of events

Match Outcome uncertainty

- Filter the data, by taking seasons starting from 2009, major leagues and exclude NA's
- The variables H,D,A are bookmaker's odds from Bet365

```
data("f_data_sm")
f_2009 <- f_data_sm %>%
  filter(SEASON > 2008,
         COUNTRY %in% c("Italy", "Germany", "Spain","England",
            "Netherlands", "Portugal"), !is.na(H) )
head(f_2009)
```

```
##       SEASON COUNTRY         LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 50107   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 50108   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 50109   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 50110   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 50111   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 50112   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##       FTHG FTAG FTTG    H   D     A
## 50107    1    0    1 1.20 6.5 15.00
## 50108    3    1    4 1.83 3.5  4.50
## 50109    2    3    5 2.00 3.3  4.00
## 50110    2    1    3 2.60 3.2  2.80
## 50111    2    1    3 3.20 3.4  2.25
## 50112    0    1    1 5.50 3.6  1.67
```

Match Outcome uncertainty

- We need to convert odds to prboabilities
- We also need to have vig ("vigorish") free probabilities, thus probabilities that sums to 1.
- Try library odds.converter

```
( x <- odds.fv(c(1.4,3.8,8), input = 'dec', output = 'prob'))
```

```
## [1] 0.6479113 0.2387042 0.1133845
```

```
sum(x)
```

```
## [1] 1
```

Match Outcome uncertainty

Replace variables H,D,A with vig-free probabilities

```
f_2009[,c("H", "D", "A")] <- odds.fv(f_2009[,c("H", "D", "A")],
                      input = 'dec',output = 'prob')
head(f_2009)
```

```
##       SEASON COUNTRY          LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 50107   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 50108   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 50109   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 50110   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 50111   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 50112   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##       FTHG FTAG FTTG         H         D         A
## 50107    1    0    1 0.7907543 0.1459854 0.06326034
## 50108    3    1    4 0.5182626 0.2709773 0.21076012
## 50109    2    3    5 0.4748201 0.2877698 0.23741007
## 50110    2    1    3 0.3648208 0.2964169 0.33876221
## 50111    2    1    3 0.2973183 0.2798290 0.42285270
## 50112    0    1    1 0.1717862 0.2624511 0.56576278
```

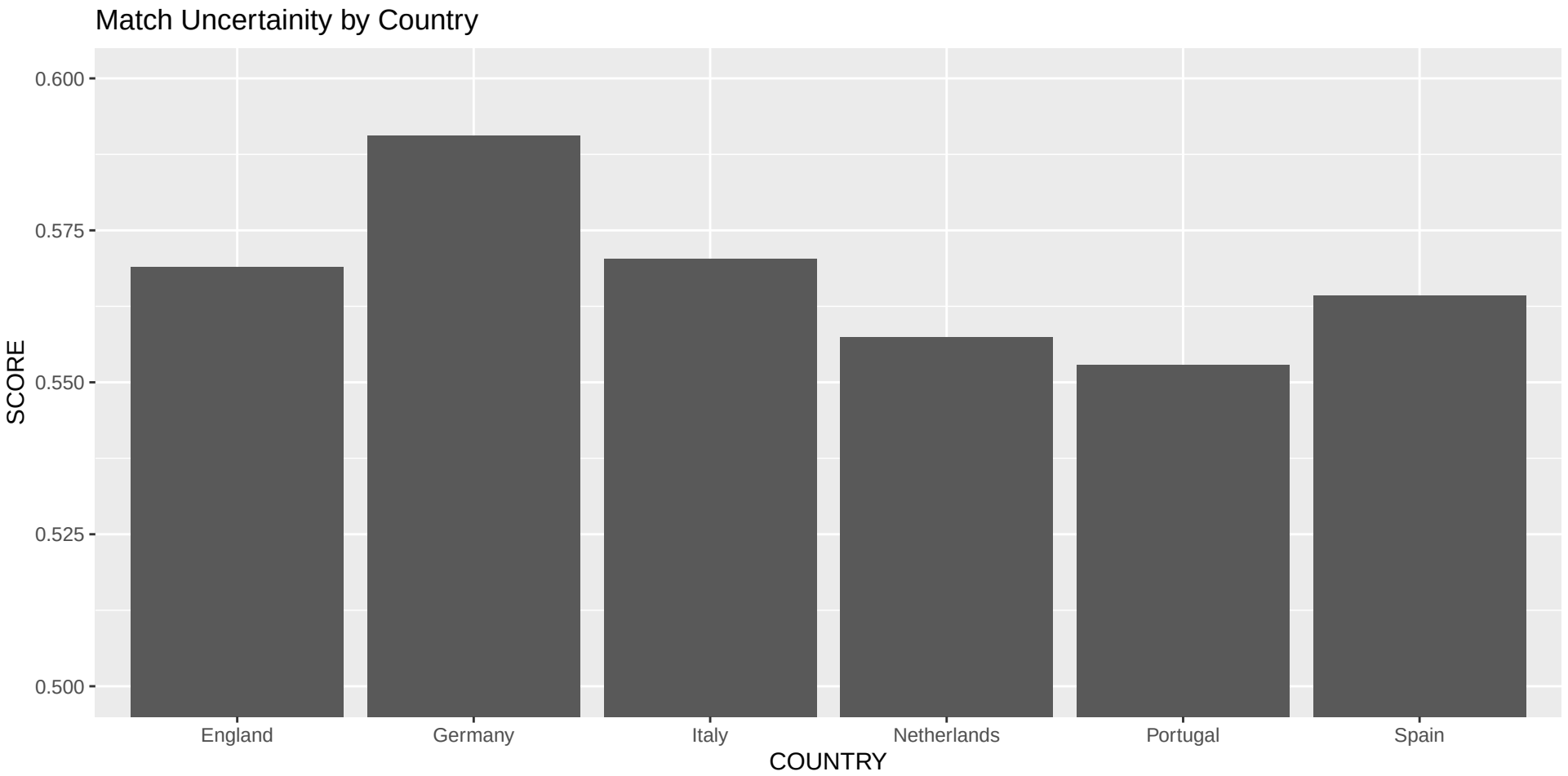
Match Outcome uncertainty

Calculate brier score by country

```
brier_score <- f_2009 %>% group_by(COUNTRY) %>%
  do(SCORE = multiclass.Brier(.[,c("H", "D", "A")], .$FTR))
brier_score <- as.data.frame(brier_score)
brier_score$SCORE <- unlist(brier_score$SCORE)
```

Match Outcome uncertainty

```
ggplot(brier_score, aes(x=COUNTRY, y = SCORE))+geom_bar(stat='identity')+
  coord_cartesian(ylim = c(0.5,0.6))+ggtitle('Match Uncertainity by Country')
```

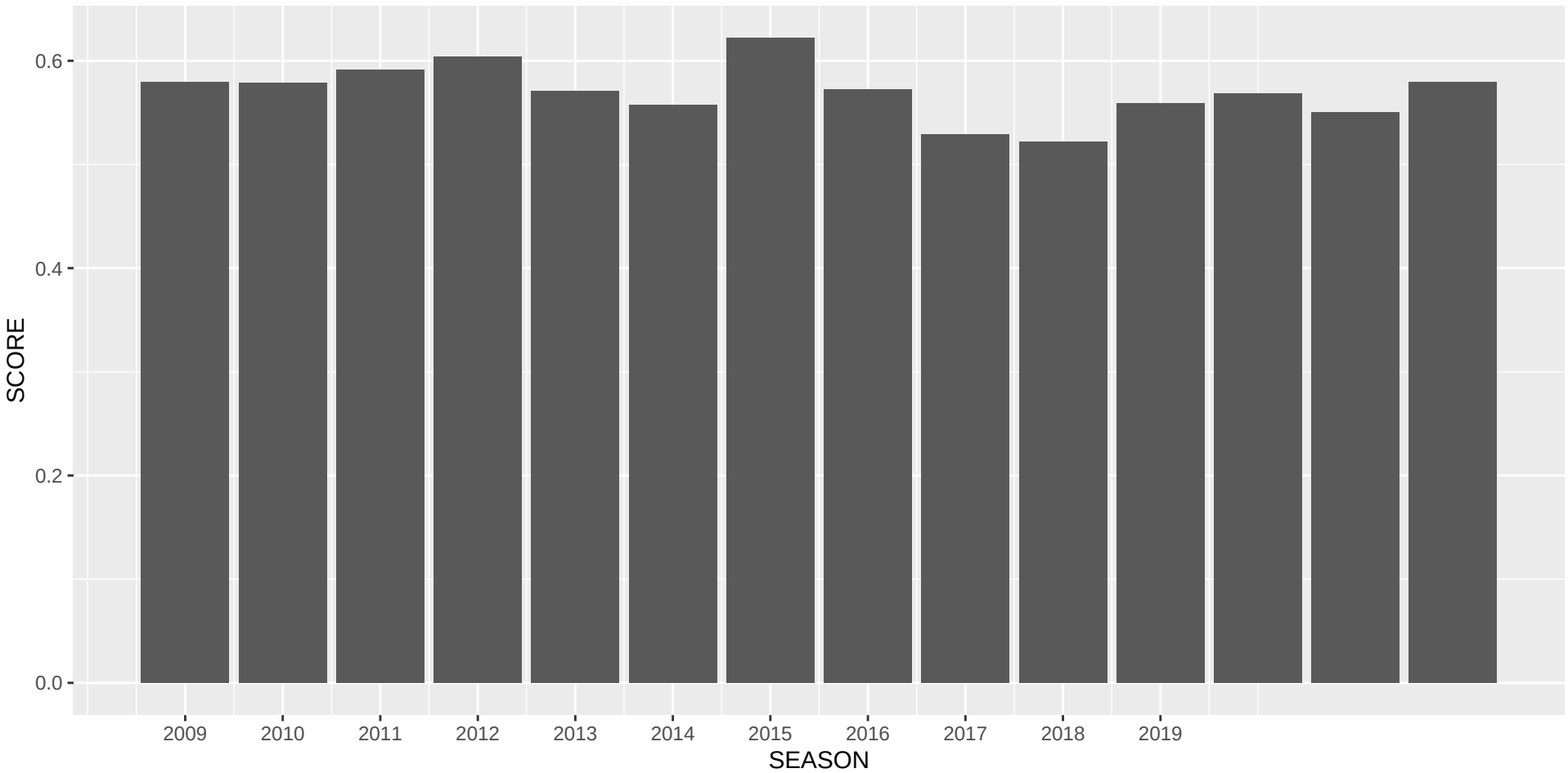


Match Outcome uncertainty

```
brier_score <- f_2009 %>% group_by(COUNTRY, SEASON) %>%
  do(SCORE = multiclass.Brier(.[,c("H", "D", "A")], .$FTR))
brier_score <- as.data.frame(brier_score)
brier_score$SCORE <- unlist(brier_score$SCORE)
```

Match Outcome uncertainty: Italy

```
brier_score %>% filter(COUNTRY=='Italy') %>%
ggplot(aes(x=SEASON, y=SCORE))+geom_bar(stat='identity') +
  scale_x_continuous(breaks = 2009:2019)
```



Match Outcome uncertainty

```
ggplot(brier_score, aes(x=SEASON, y=SCORE))+
  geom_bar(stat='identity',position=position_dodge(width=1))+
  facet_grid(~COUNTRY)+ coord_flip()+ggtitle("Brier Score by Country")
```

